

Engineering Reboot 2026

Day 25 Engineering Workbook

Version 1.0 – Implementation Edition (Locked)

Mission

Extend the Employee Directory by persisting employee data using JSON so information survives between program executions.

Learning Objectives

Explain persistence.

Serialize Python objects to JSON.

Deserialize JSON into Python objects.

Update stored data.

Separate business logic from storage.

Engineering Theory

Persistence preserves application state. JSON maps naturally to Python dictionaries and lists. Serialization writes data; deserialization reads it back.

Week 5 Engineering Scenario

Replace the hardcoded employee dataset with employees.json and maintain employee information between executions.

Reference Dataset

employees.json created from the canonical EMPLOYEES dataset introduced on Day 23.

Exercise 1

json_write_demo.py: Write EMPLOYEES to employees.json using `json.dump(..., indent=4)`.

Exercise 2

json_read_demo.py: Load employees.json and display all employees.

Exercise 3

json_update_demo.py: Increase employee ID 1005 salary by 5%, save the file, and verify persistence.

Mini Project

Persistent Employee Directory supporting load, display, search, add, update, delete, save and exit.

Stretch Goals

Automatic backups; timestamps; department export; pretty-printed JSON; duplicate ID prevention.

Engineering Checklist

JSON loads and saves successfully; data persists after restart; unique IDs maintained; reusable functions.

Git Commit

Day 25 - Completed persistent employee directory using JSON

Engineering Takeaways

Understand persistence, JSON, serialization, deserialization, and separation of storage from business logic.